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Lab 1: Implementation of Source code management using git commands through GitHub.

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OBJECTIVE

Implementation of Source Code Management Using Git Commands Through GitHub.

AIM

To understand and implement Source Code Management (SCM) using Git and GitHub for tracking changes, managing versions, and collaborating on software projects.

THEORY

Source Code Management (SCM) is a software development practice used to manage and monitor changes made to source code over time. It allows developers to maintain different versions of a project, track modifications, collaborate efficiently, and restore earlier versions whenever required.

Git is a distributed version control system that records changes made to files and enables multiple developers to work on the same project simultaneously without affecting each other's work. It maintains a complete history of all project revisions, making version management simple and reliable.

GitHub is a web-based platform that hosts Git repositories online. In addition to version control, it offers collaboration features such as pull requests, issue tracking, repository sharing, and project management, making it easier for teams to develop software together.

Together, Git and GitHub play a significant role in modern software development, especially in Agile environments. They support efficient collaboration, version tracking, continuous integration, and code review, helping development teams build and maintain software more effectively.

OBJECTIVES

- Learn the basics of Git.
- Create and manage Git repositories.
- Track project changes using commits.
- Connect local repositories with GitHub.
- Push and pull project updates.
- Understand collaborative software development.

ALGORITHM

1. Install Git on the system.
2. Create a GitHub account.
3. Create a repository on GitHub.
4. Initialize a local Git repository.
5. Add project files.
6. Stage files using Git.
7. Commit changes.
8. Link local repository to GitHub.
9. Push files to GitHub.
10. Verify successful synchronization.

CODE

Initialize Repository

```
git init
```

Configure User Information

```
git config --global user.name "your name"  
git config --global user.email "your@email.com"
```

Check Repository Status

```
git status
```

Add Files

```
git add .
```

Commit Changes

```
git commit -m "Initial Commit"
```

Connect Repository to GitHub

```
git remote add origin https://github.com/username/repository.git
```

Push to GitHub

```
git branch -M main  
git push -u origin main
```

Pull Latest Changes

```
git pull origin main
```

OUTPUT

The Git repository was successfully initialized and connected to GitHub.

Repository created successfully.

Files tracked using Git.

Changes committed successfully.

Remote repository linked.

Project pushed to GitHub.

RESULT

The implementation of Source Code Management using Git and GitHub was completed successfully. Version control operations such as repository creation, staging, committing, and synchronization with GitHub were performed successfully.

CONCLUSION

This experiment provided practical knowledge of Source Code Management using Git and GitHub. It demonstrated how version control systems help manage software projects efficiently by maintaining code history, enabling collaboration, and supporting Agile Software Development practices.